

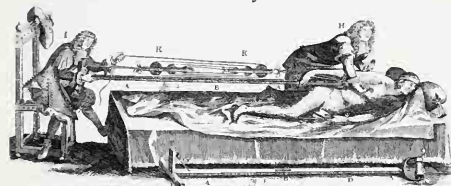
# Inside bones



## GROWING BONE

In the early embryo (p. 44), the body does not have bones. Their shapes begin as a softer substance, cartilage, or gristle. Nuggets of bone, known as ossification centers, then develop at certain points within the cartilage. They grow and spread, turning the cartilage into true bone. In this unborn baby's hand, speckled areas of bone are replacing the clear areas of cartilage.

BONES ARE MOST OFTEN SEEN as pale, dry, flaky objects in museums or as the last remnants of a dead animal lying in the countryside. They might not seem very interesting, but they do have a life of their own. Our prehistoric ancestors would have had a more intimate knowledge of bones than people today. More than a million years ago, early humans used rock axes and stone hammers to crack open animal bones, possibly to get at the nutritious marrow inside. They would have seen that a typical bone has a tough skin that we call the periosteum, covering an outer shell of compact bone, which is hard and dense. The compact bone encloses an inner layer of lighter, honeycombed cancellous or spongy bone, with soft marrow at the center. When early microscopists peered at thin slices of bone, they were amazed to see a complex and regular microarchitecture, which reflects bone's active life. The structural unit of compact bone is the tube-shaped Haversian system, named after English physician Clopton Havers (1650-1701). He published his observations in 1691, opening the way for the research of later physiologists, who found that bones had many roles in addition to holding up the body.



## SETTING BONES

Fossilized human skeletons 100,000 years old show how broken bones were repositioned to heal better. Bone setting is an ancient art. Here a late 17th-century rope-and-pulley invention is pulling broken arm bone back into place.



## SPONGY BONE

This microscopic view of cancellous, or spongy, bone shows its open system of bony lattices and struts, known as trabeculae. In living bone the spaces are filled with body fluids and wandering cells (p. 13). Landing from an energetic leap, muscles exert tremendous pressure in the hip and knee joints, so the bone must stand compression (squeezing) forces of up to half a ton (0.55 tonnes).

## BONES AS TUBES

Engineers have long known that weight-for-strength, few structural shapes rival the tube. This cutaway humerus from the upper arm shows how the main shaft is a narrow tube, while the ends are expanded to spread the pressure in their joints with adjacent bones.

## INSIDE THE HIP

In the hip joint, the ball-shaped head of the femur (thigh bone) fits into a cup-shaped socket in the pelvis (hip bone). The head of the femur is mainly cancellous, or spongy, bone surrounded by a thin shell of dense compact bone. Engineers have analyzed the stresses and strains on bones such as this and constructed mechanical models. They find that compact bone develops where needed, in the areas of most stress. Cancellous bone is a "filler." It is not so strong, but it is lighter and less of a drain on the body's mineral supplies.

